FARZANA FATHIMA K S

Software Developer | Backend Developer

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PROFESSIONAL SUMMARY

Computer Science graduate and current Python Full Stack Development Intern with experience in web applications, REST API integration, SQL databases, and cybersecurity. Skilled in Python, Java, C, Django, Flask, and object-oriented programming. Project experience includes a Flutter learning application and an AI-enabled IoT landslide monitoring system. Seeking software development and backend development opportunities.

TECHNICAL SKILLS

Programming: Python, Java, C, object-oriented programming (OOP).

Web and Application Development: Django, Flask, HTML, CSS, JavaScript, Flutter, front-end and back-end development, REST API integration, CRUD operations.

Databases and Data Analysis: SQL, database connectivity, database management systems (DBMS), Excel, Pandas, NumPy.

Computer Science: Data structures and algorithms, algorithm design, problem solving.

AI and Machine Learning: Artificial intelligence, machine learning, deep learning, Random Forest, autoencoders, Bayesian techniques.

IoT and Tools: ESP32, environmental sensors, ThingSpeak, Streamlit, Git, VS Code.

Testing and Security: Debugging, software testing, vulnerability assessment, penetration testing exercises, security incident analysis, security policies.

PROJECTS

CompGenio

- Developed a Flutter application for KTU Semester 6 students to practise Comprehensive Course Work examinations.
- Created an interactive, user-friendly interface to support learning and exam preparation.

Landslide-X

Early Landslide Detection Using IoT, Machine Learning and Deep Learning

- Developed an AI-IoT landslide monitoring system using ESP32, environmental sensors, and ThingSpeak for real-time data collection and monitoring.
- Implemented Random Forest, autoencoder, and Bayesian techniques to improve prediction accuracy and reduce false alarms.
- Designed a Streamlit dashboard with automated email alerts for early warning in landslide-prone areas.

INTERNSHIP EXPERIENCE

Python Full Stack Development Intern

SMEC Technologies Pvt. Ltd. - 2026 - Present

- Develop web applications using Python, Django/Flask, HTML, CSS, JavaScript, and SQL.
- Work on front-end and back-end development, including REST API integration and database connectivity.
- Implement CRUD operations and manage SQL databases.
- Practise debugging, testing, Git version control, and collaborative software development.
- Apply Python and OOP concepts to develop and maintain application features.

Cybersecurity Intern

Retechnox Technologies - November 2024 - December 2024

- Gained hands-on experience in HTML, CSS, and basic JavaScript for building web pages.
- Developed and implemented cybersecurity protocols to strengthen data protection.
- Analysed security incidents, identified vulnerabilities, and refined incident response strategies.
- Assisted with security policies for mobile devices and remote access to safeguard sensitive data.
- Participated in penetration testing and simulated cyberattack exercises to assess security measures and response preparedness.
- Contributed to end-user security awareness training and evaluated cybersecurity tools and technologies.
- Collaborated with the IT team to identify security risks and develop mitigation strategies.

EDUCATION

Bachelor of Technology in Computer Science

College of Engineering Pathanapuram, Kollam - April 2026

PROFESSIONAL SKILLS

Communication, teamwork and collaboration, leadership, presentation skills, problem solving, and consistent performance.

LANGUAGES

English, Hindi, Malayalam

INTERESTS

Web development

Software testing

Learning programming languages

Reading technology blogs and articles

Volunteering